

Connections

➡ Building on

- The rest of the book [ch1–ch42](#)

Arriving at

- How organic chemistry produced an AIDS treatment in collaboration with biologists
- How organic chemists are in the front line of the fight against epidemics
- Where organic chemistry might be going next

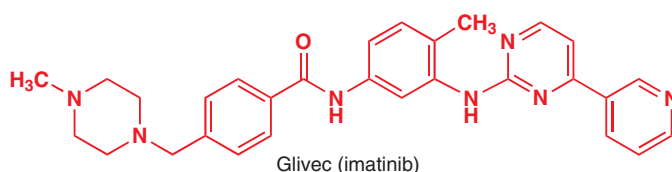
➡ Looking forward to

- Life as a chemist

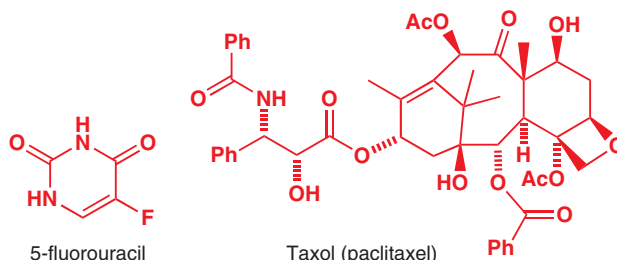
Science advances through interaction between disciplines

Just as the stonemasons who worked out how to build the great gothic cathedrals of the middle ages transformed architecture, the organic chemists who build molecules on a scale 10^{10} times smaller have transformed our expectations of everyday life. When we are ill, we expect there to be a drug to treat us; when we need to mend something, we expect there to be a sealant, glue, or coating to solve our problem. We expect paints, plastics, and clothes of any colour. If an engineer needs a material with certain properties of strength and flexibility, she expects organic chemists to be able to make it. If a biologist needs a molecule to inhibit an enzyme selectively, he expects organic chemists to be able to make it. In the future the same will be true of plastics that conduct electricity or emit light, or drugs tailored to your own individual genetic makeup. The creative art of organic chemistry has transformed our ability to understand and manipulate the world on a molecular scale and above, and it has been able to do this because of the collaboration between those who can make molecules and those who can use them—between chemists, physicists, engineers, and materials scientists.

The most dramatic scientific developments involving organic chemistry at the beginning of the 21st century are new methods in medicine from collaborations between organic chemists and biologists. Progress is slow but secure across a whole range of diseases long thought impregnable to treatment. That media favourite, ‘the cure for cancer’, is already not just ‘a cure’ but hundreds of successful cures for the hundreds of diseases collectively called ‘cancer’. At the turn of the twenty-first century, it was the case that there was *some* chance of survival for all known types of childhood cancer. The drug Glivec, launched in 2001, now essentially cures 75% of patients with chronic myeloid leukemia. 5-Fluorouracil is a well-established chemotherapy drug that slows down the progression of cancer. But in conjunction with Avastin, which prevents tumours developing their own blood supply, it is much more effective against certain colon cancers. Avastin in conjunction with Taxol (launched in 1992) increases Taxol’s effectiveness against breast cancer. Avastin was launched in 2004, and is expected to be the world’s biggest selling drug by 2014.



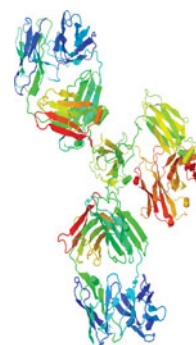
Glivec (imatinib)



5-fluorouracil

Taxol (paclitaxel)

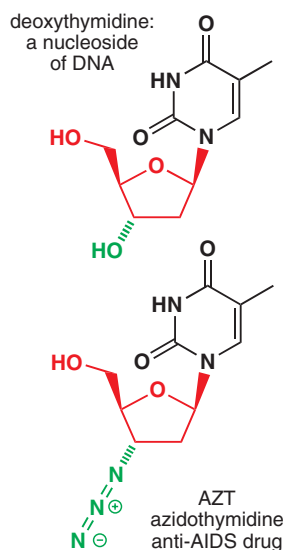
Ph



Avastin (bevacizumab)

5-Fluorouracil could hardly be simpler: it interferes with cell proliferation by modifying natural uracil to incorporate a stubbornly unreactive fluorine. Taxol is a rare metabolite of the Pacific yew tree that can be made at great expense in the laboratory, and for a while was produced by chemical modification of a common precursor that can be harvested. It is now made by fermentation using cultured yew tree cells. Avastin is at the other end of the scale of complexity: it is an antibody against a protein involved in blood vessel growth, and we have represented only its gross structure: a detailed structural diagram would be huge. The antibody was induced in mice, its protein sequence was determined and then modified using the techniques of molecular biology which grew out of organic chemistry in the 1960s and 1970s, and it is produced by expression of the modified gene in bacteria. Which of this is chemistry, which is biology, and which is medicine? There is no point deciding.

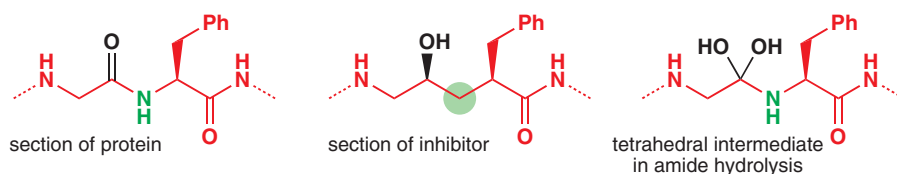
Chemistry vs viruses



We are going to spend most of this chapter discussing two medical developments, both battles pitting chemists against viruses: one is partly won, and one has fortunately not yet been fought. Like cancer, viruses are an insidious menace because they subvert the body's own biochemical machinery to cause harm, but since the middle of the last century, with antibiotics being used to treat bacterial infections, the threat from infectious disease seemed to be in retreat. So when AIDS (acquired immune deficiency syndrome) first came into the news in the 1980s, medics struggled to explain the mysterious deaths from normally harmless diseases after the patient's immune system had been weakened and eventually destroyed. But the cause was soon identified by biologists as a new virus, HIV (human immunodeficiency virus), and antiviral drugs, notably AZT, were used with some success. These drugs imitate natural nucleosides (AZT imitates deoxythymidine) and inhibit the virus from copying its RNA into DNA inside human cells by inhibiting the reverse transcriptase enzyme.

As is often the problem with antiviral (and anticancer) chemotherapy, the drugs also inhibit the normal function of essential human enzymes and are very toxic. But biologists discovered an alternative point of attack. An enzyme unique to the virus cuts up long proteins into small pieces essential for the formation of new HIV particles. If this enzyme could be inhibited, no new viruses would be formed and neither should the inhibitor interfere with human biochemistry.

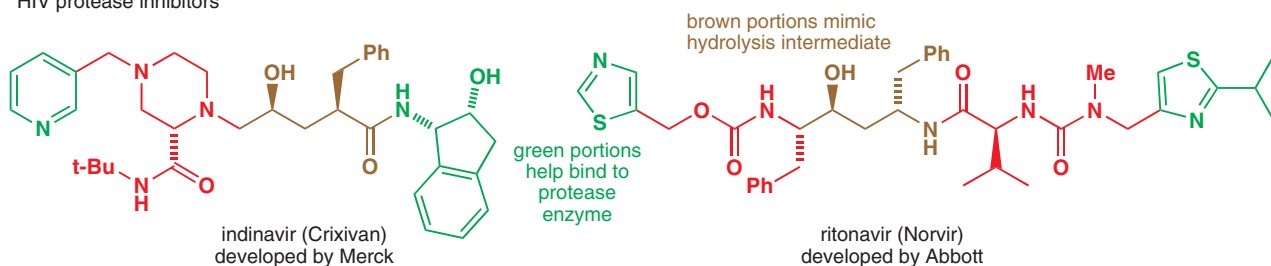
Blocking HIV protease inhibitors means mimicking the proteins they slice up, but real peptides are usually poor drugs because humans have their own peptidases which quickly cut up ingested proteins by hydrolysis of the amide link. The solution is to make a drug which looks like the peptide but can't be hydrolysed because the C–N bond of the peptide has been replaced by a C–C bond (green parts of the structures below).



This stops the drug being hydrolysed, but the drug also has to stop the viral protein being hydrolysed. To get it to do this, medicinal chemists used another trick. Enzymes work by binding the transition state for a reaction, and while of course the chemists couldn't make a transition state (it is by its nature unstable) they made a molecule with a sufficient resemblance to the tetrahedral intermediate for amide hydrolysis (black parts of molecules above) that the protease is tricked into taking it into its active site, where it blocks the protease's function.

The knowledge that only one of the two hydroxyl groups of the tetrahedral intermediate was needed was acquired from an X-ray crystal structure showing how the enzyme binds the substrate. Other structural information was also used to design the drugs: for example, HIV protease is a dimeric enzyme and experience with this class of protease suggested correctly that more or less symmetrically placed aromatic or heterocyclic rings would greatly improve binding. Two successful protease inhibitors are shown below, with the active site binding portion in brown and the heterocyclic binding portions in green.

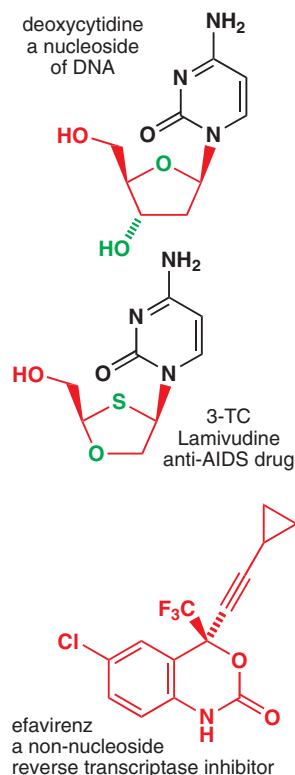
HIV protease inhibitors



These developments looked so promising that Merck set up a new research station at West Point, Pennsylvania, dedicated to this work. The biochemist in charge, Dr Irving Sigal, was one of the victims of the Lockerbie bombing in 1988 but his work lived on in Crixivan (indinavir). In combination with the antiviral agents AZT and 3TC (Lamivudine), shown with the nucleoside it imitates, indinavir revolutionized the treatment of HIV in the 1990s. Before the use of 'combination therapy', as it is known, most of those with HIV were dead within 2 years. Now no-one knows how long they will survive as the combination of the three drugs reduces the amount of virus to undetectably low levels.

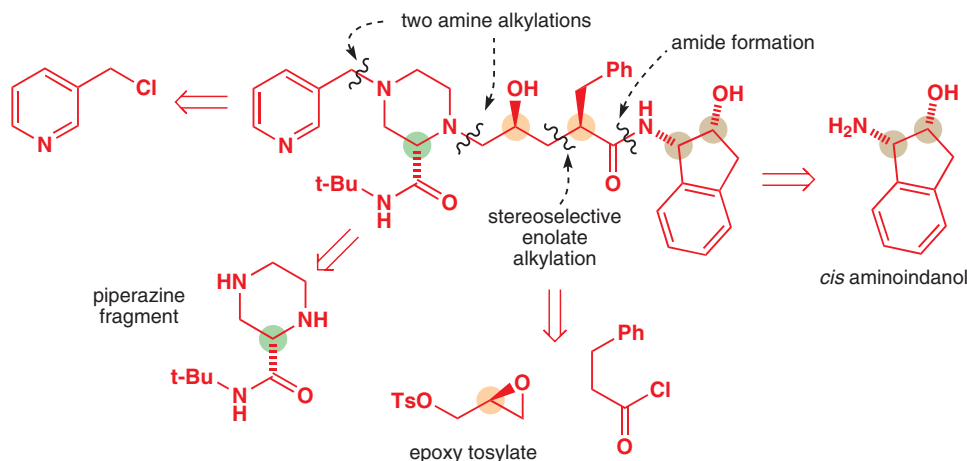
The AIDS crisis led to cooperation between the pharmaceutical companies unparalleled since the development of penicillin during the Second World War. Fifteen companies set up an AIDS drug development collaboration programme, with government agencies and universities contributing as well. The battle is not yet won, of course, and the HIV protease inhibitors have now been joined by a new generation of non-nucleoside reverse transcriptase inhibitors, such as the DuPont–Merck compound efavirenz. These commonly join the other drugs of the types mentioned above as part of the drug regimes known as 'highly active antiretroviral therapy' or HAART. The mixture of drugs used to combat HIV changes as discoveries are made, but life-saving combination therapy of this sort would not be possible without the sort of collaboration between organic chemists, biochemists, virologists, X-ray crystallographers, and molecular modellers that went into discovering and making indinavir.

After indinavir was found to be effective, the job of the chemists was an exceptionally urgent task. They knew that a kilo of compound was needed to keep each patient alive and well for a year (newer HIV protease inhibitors require much smaller doses). Merck built a dedicated plant for the manufacture of Crixivan at Elkton, Virginia, in 1995. Within a year, production was running at full blast and there are millions of people alive today as a result.



The synthesis of indinavir

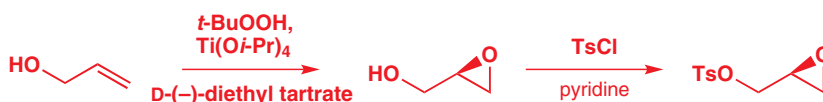
Indinavir was a formidable synthetic target. It was probably the most complex compound ever made in quantity by organic synthesis and the 3 g per day dose meant that huge quantities were required. The complexity largely arises from the stereochemistry. As with all chiral new drugs, it is a single enantiomer: there are five stereogenic centres, marked with coloured circles on the diagram below, and their disposition means that three separate pieces of asymmetric synthesis must be devised.



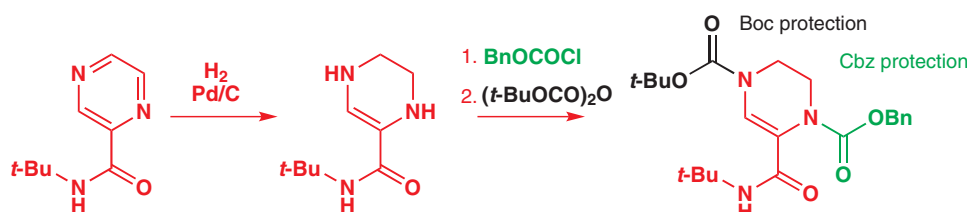
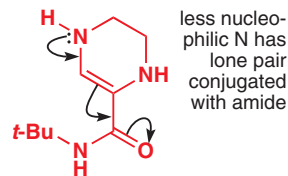
The challenge with indinavir, as with any drug, is to make it efficiently: high yields, few steps. We can start by looking at some likely disconnections, summarized in the scheme above. They are all disconnections of the sorts you met in Chapter 28, and they all correspond to reliable reactions. These disconnections split the molecule into five manageable fragments, three of which contain stereogenic centres and will have to be made as single enantiomers. One of the orange stereogenic centres would have to be made in the enolate alkylation step, so this step will need to be diastereoselective.

Let's take the three chiral fragments in turn. First, the simplest one: the central epoxide. The reagent we need here will carry a leaving group, such as a tosylate, to allow it to alkylate the piperazine to the left, and it can easily be made from an epoxyalcohol. This gives a very good way of making this compound as a single enantiomer—a Sharpless asymmetric epoxidation of allyl alcohol.

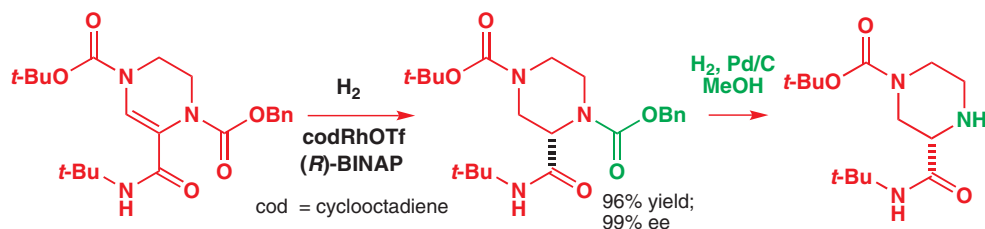
➡ Sharpless asymmetric epoxidation was discussed in Chapter 41, p. 1120.



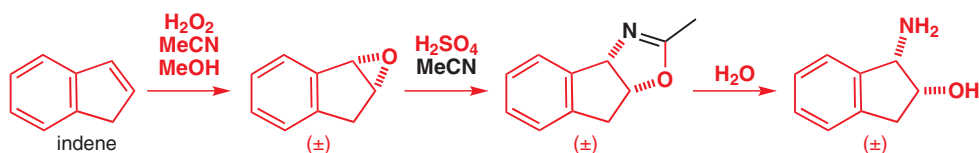
Next, the piperazine fragment. This has two nucleophilic nitrogen atoms and they will both need protecting with different protecting groups to allow them to be revealed one at a time. It will also need to be made as a single enantiomer. In an early route to indinavir, this was done by resolution, but enantioselective hydrogenation provides a better alternative. Starting from a pyrazine derivative, a normal hydrogenation over palladium on charcoal could be stopped at the tetrahydropyrazine stage. The two nitrogens in this compound have different reactivities because one is conjugated with the amide while one is not (the curly arrows in the margin show this). The more nucleophilic nitrogen—the one *not* conjugated with the amide—was protected with benzyl chloroformate to give the Cbz derivative. Now the less reactive nitrogen can be protected with a Boc group, using DMAP as a base.



You met asymmetric hydrogenation using BINAP complexes of rhodium in Chapter 41 as a method for the synthesis of amino acids. The substrate and catalyst are slightly different here, but the principle is the same: the chiral ligand, BINAP, directs addition of hydrogen across one of the enantiotopic faces of the double bond with almost perfect enantioselectivity and in very high yield. A further hydrogenation step allowed selective removal of the Cbz group, preparing one of the two nitrogen atoms for alkylation.

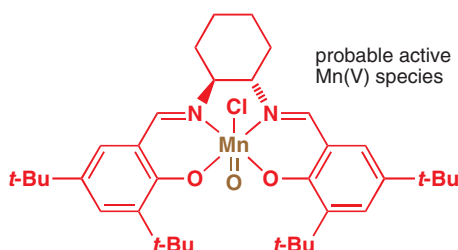


The remaining chiral fragment is a compound whose synthesis was discussed in Chapter 39. It can be made on a reasonably large scale (600 kg) in one reaction vessel, starting from indene. First, the double bond is epoxidized, not with *m*-CPBA but with the cheaper hydrogen peroxide in an acetonitrile/methanol mixture, which generates a peroxyimide acid (the C=N analogue of a peracid) as the active oxidant. Acid-catalysed opening of the epoxide leads to a cation, which takes part in a reversible Ritter reaction with the acetonitrile solvent, leading to a single diastereoisomer of a heterocyclic intermediate, which is hydrolysed to the amino-alcohol.



The product is, of course, racemic but, as it is an amine, resolution with an acid should be straightforward. Crystallization of its tartrate salt, for example, leads to the required single enantiomer in 99.9% ee. With such cheap starting materials, resolution is just about acceptable, even though it wastes half the material. It would be better to oxidize the indene enantioselectively, and the solution here, as you saw in Chapter 41, is to use a Jacobsen epoxidation, which gives the epoxide in 79% yield and 84% ee.

Jacobsen asymmetric epoxidation of indene



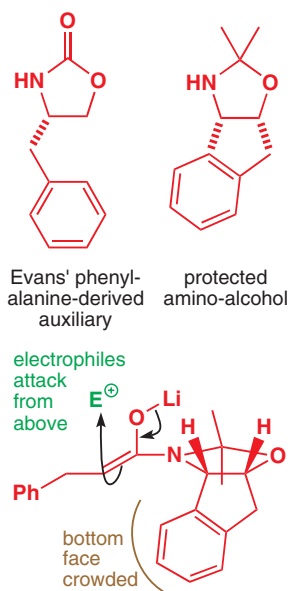
Only one, orange, stereogenic centre remains, and its stereoselective formation turns out to be the most remarkable reaction of the whole synthesis. The centre is the one created in the planned enolate alkylation step, shown in the margin. The obvious way to make this centre is to make Y a chiral auxiliary, which would direct a diastereoselective alkylation before being removed and replaced with the amino-alcohol portion.

But the Merck chemists noticed that amino alcohol itself, certainly once protected, has a remarkable similarity to Evans' oxazolidinone auxiliaries anyway, and it turns out that this amino alcohol will function very successfully as a chiral auxiliary, which does not need to be removed, avoiding waste and saving steps! The amino alcohol was acylated with the acyl chloride, and the amide was protected as the nitrogen analogue of an acetonide by treating with 2-methoxypropene (the methyl enol ether of acetone) and an acid catalyst. The enolate

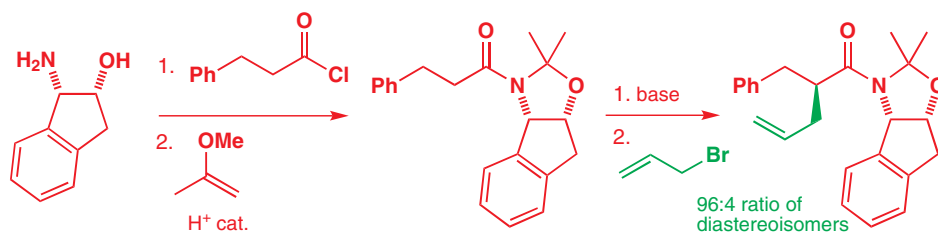
➡ Turn to pp. 1066–1067 for details of the mechanisms in this reaction sequence and an explanation for its *cis* diastereoselectivity.



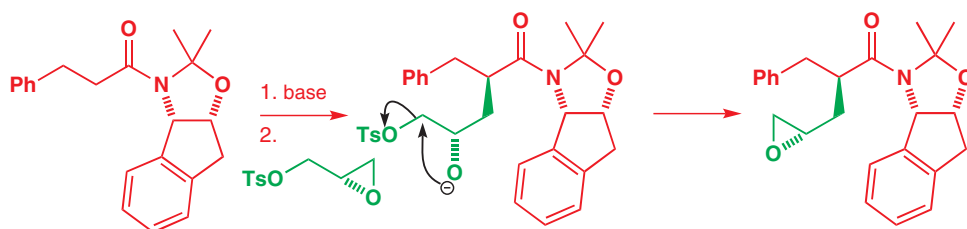
■ Evans auxiliary-directed alkylation is described in Chapter 41, p. 1109.



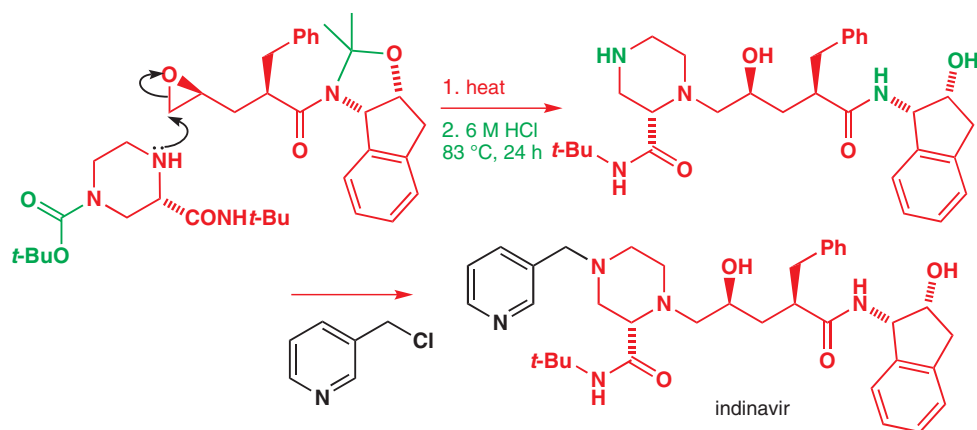
of this amide reacts highly diastereoselectively with alkylating agents, including, for example, allyl bromide.



The reason for the stereoselectivity is not altogether clear, but we would expect the bulky nitrogen substituents to favour formation of the *cis* enolate. With the amino-alcohol portion arranged as shown, the top face is more open to attack by electrophiles. The enolate also reacts diastereoselectively with the epoxy-tosylate prepared earlier. The epoxide, being more electrophilic than the tosylate, is opened first, giving an alkoxide, which closes again to give a new epoxide. The absolute configuration at the stereogenic centre within the epoxide was, of course, already fixed (by the earlier enantioselective Sharpless epoxidation).



Three of the five fragments have now been assembled, and only the two amine alkylations remain. The first alkylation makes use of the epoxide to introduce the required 1,2-amino-alcohol functionality. The protected enantiomerically pure piperazine reacted with the epoxide, and the product was treated with acid to deprotect both the second piperazine nitrogen and the *gem*-dimethyl group left over from the earlier chiral auxiliary step. The newly liberated secondary amine was alkylated with the reactive electrophile 3-chloromethyl pyridine, and the final product was crystallized as its sulfate salt.



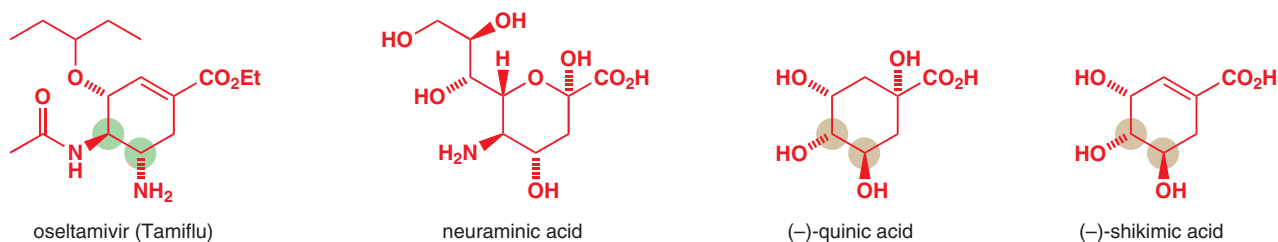
The synthesis of oseltamivir

Our second example of the use of chemistry to save lives is more recent. Several times in the last century major epidemics of influenza have caused deaths, sometimes running into millions. Virologists tell us that a global influenza pandemic is a constant danger, and a number of times in recent years highly aggressive forms of the flu virus have found their way from other animals (often poultry or pigs) into the human population. Fortunately, at the time of writing, none has caused more than a few thousand deaths, the most serious being the swine flu pandemic of 2009–10, which claimed the lives of 18,000 people, many of them in Mexico.

To put this in context, the 1918 flu epidemic, which was caused by the same strain (H1N1) of virus, killed 50–100 million, 3% of the world's population at the time.

Vaccination can prevent the spread of flu, but influenza vaccines are slow to produce and difficult to generalize because of the rate of mutation of the virus. So the first line of defence is a class of antiviral compounds known as neuraminidase inhibitors. Neuraminidase is an enzyme used by the flu virus that targets human cell-surface carbohydrates containing neuraminic acids and allows the virus to release itself from the host cell. Inhibition of this enzyme prevents the new virus particles from spreading.

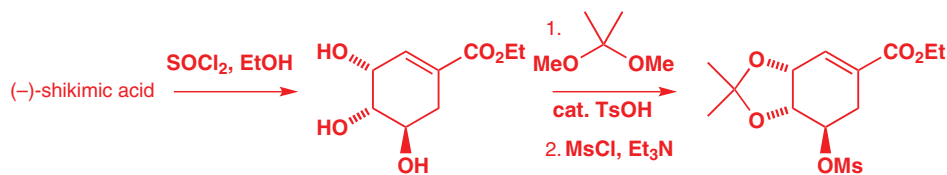
The drug oseltamivir (Tamiflu), developed by the companies Gilead and Roche, is a neuraminidase inhibitor. Like the HIV proteases described above, it has enough structural similarity with the enzyme's substrate to bind to the enzyme, but once bound it blocks the enzyme's activity. No-one knows how much oseltamivir might be needed if ever a flu pandemic took hold, but clearly the safest course of action is to stockpile the compound in readiness for such an event. The first manufacturing route to oseltamivir made use of the natural product (–)-quinic acid as a naturally derived starting material. Quinic acid is found in coffee beans, but is not available in sufficient quantities for widespread use.



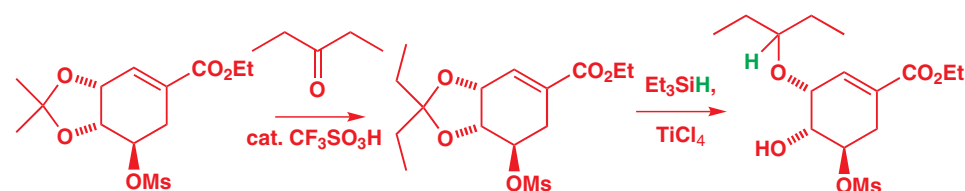
A preferable starting material, and the one that for several years now has been used as the source of the commercial drug, is (–)-shikimic acid. Shikimic acid is the plant metabolite that provides the biochemical precursor to the aromatic amino acids such as phenylalanine, tyrosine, and tryptophan. It is abundant in the spice star anise, grown in China, which can yield 3–7% of shikimic acid.

The similarity of both quinic and shikimic acid with the target drug is obvious; what is perhaps remarkable is just how many steps it takes to get from one to the other. The majority of these steps are concerned with the introduction of the two amino substituents with inversion of stereochemistry at the coloured stereogenic centres. Chiral pool syntheses often have to take long convoluted routes to correct relatively minor 'errors' of structure and stereochemistry. Here this is simply the price we have to pay for a starting material that has the valuable qualities of enantiomeric purity and the right hydrocarbon skeleton.

Oseltamivir is an ethyl ester, and esterification comes first, followed by selective protection of the *cis* diol (the *cis*-6,5-ring system is more stable than the alternative *trans*) and conversion of the remaining hydroxyl group to a methanesulfonate leaving group.

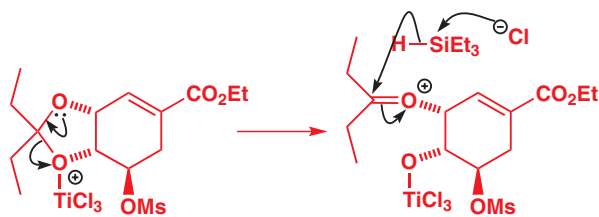


The dioxolane, which is crystalline and easily purified, is then exchanged for the acetal derived from pentan-3-one, ready for a reduction to the rather challenging hindered ether (direct alkylation with a hindered alkyl halide would struggle to avoid competing E2 elimination).

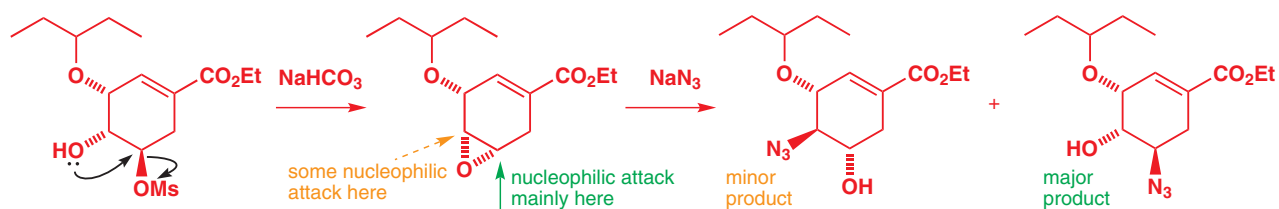


➡ You met shikimic acid in Chapter 42, p. 1154.

The reduction of the acetal is catalysed by a Lewis acid and goes via an oxonium ion, which collects hydride from the mild reducing agent triethylsilane. Silanes react only with cationic electrophiles. The oxonium ion could open either way, but this one is less hindered and possibly allows the titanium some favourable interaction with the mesylate substituent.

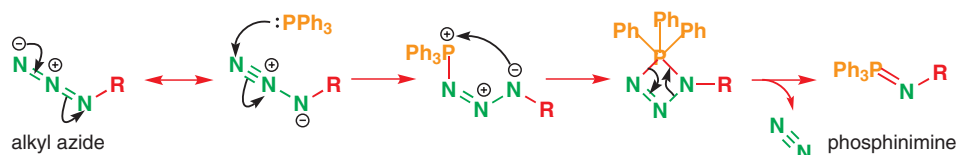


As often in the synthesis of 1,2-difunctionalized compounds, an epoxide is a key intermediate, and in this case an epoxide forms by closure of the newly revealed hydroxyl group onto the mesylate leaving group in base.

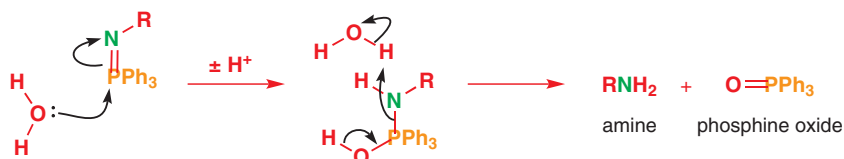


Two amino groups now need introducing with rather specific stereoselectivity, and the next key intermediate is an aziridine, the nitrogen analogue of an epoxide. Azide is not completely regioselective in opening this epoxide, but both regioisomers are formed with complete inversion of configuration.

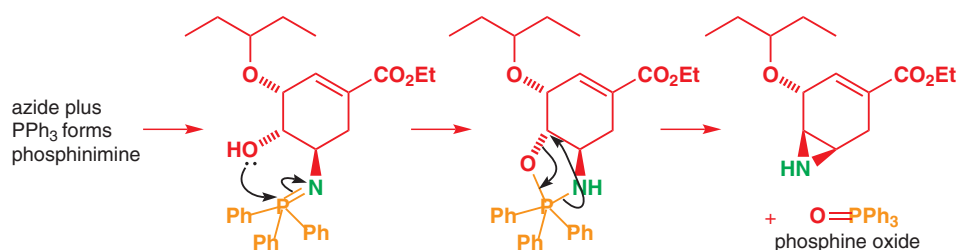
Azides may be reduced to amines with triphenylphosphine in what is known as the Staudinger reaction. The probable mechanism involves attack of triphenylphosphine on the azide and formation of a phosphinimine via a four-membered intermediate—notice the similarity with the Wittig reaction!



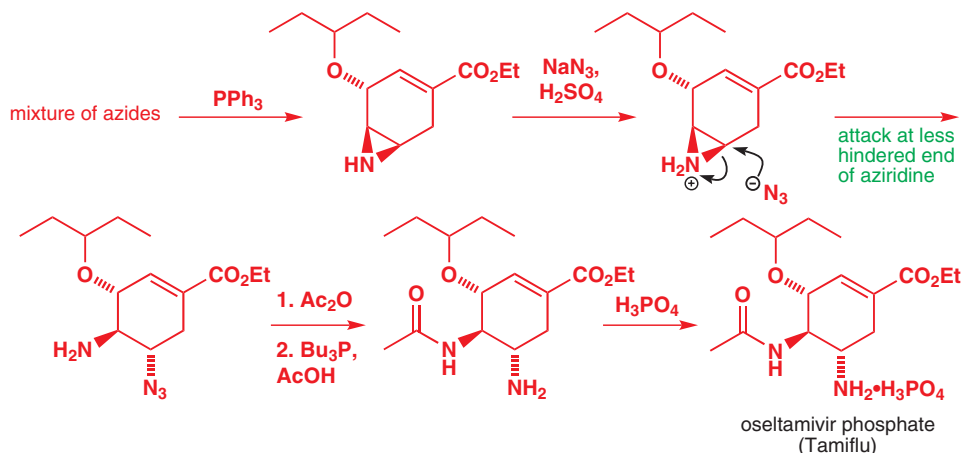
The phosphinimine, in the presence of water, hydrolyses to an amine—overall a molecule of nitrogen is lost and a molecule of water is 'dismembered' and shared between the reagents.



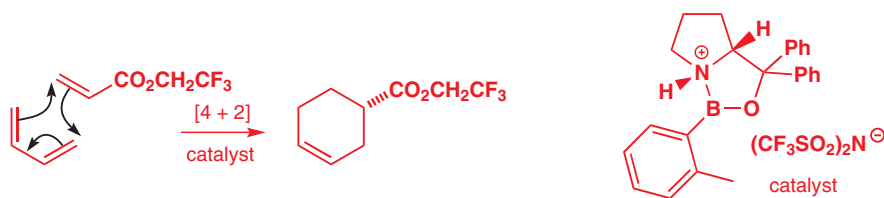
When the azide has an adjacent hydroxyl group, something more interesting happens: the phosphinimine intermediate in the reaction is intercepted by the alcohol, which turns itself into a leaving group. Extrusion of the stable phosphine oxide gives an aziridine, with inversion of stereochemistry as the nitrogen displaces the leaving group. Here is the result with the major azide from the oseltamivir synthesis:



In this case, it doesn't matter which azide you start with: triphenylphosphine converts them both to the same aziridine. Like epoxides, aziridines open with nucleophiles under acid catalysis, and azide is used again to put in the second amino group by attack at the less hindered end of the aziridine. To get the right amino group acetylated, the amide is formed before the azide is reduced, this time with tributylphosphine. The drug is formulated as a stable phosphate salt by treatment with phosphoric acid.



This is not, by any stretch of the imagination, an efficient synthesis, not least because there are two uses of potentially explosive azides, and large amounts of waste are produced from the phosphine steps. However, for several years it was the best route available, and Roche operated it as a manufacturing process on a tonne scale. In the last few years, however, several modifications have been published, and among the most efficient of the alternatives was one devised in 2006 by the Nobel prize-winning chemist E. J. Corey. Corey's route built on the fact that oseltamivir is a cyclohexene, and as you saw in Chapter 34 cyclohexenes are made efficiently by a Diels–Alder reaction.

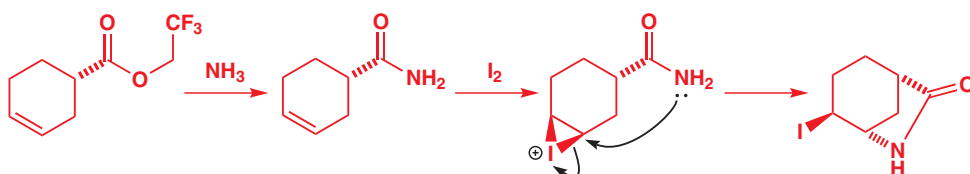


Corey's research group combined the two very cheap reagents butadiene and trifluoroethyl acrylate in the first step of their alternative synthesis: the cycloadduct already has the scaffold of oseltamivir. Not starting with a natural product has its advantages and disadvantages: no longer is supply limited by the world production of coffee beans or star anise, and no longer is there a need to make do with a compound of the wrong relative stereochemistry, wasting valuable resources inverting stereogenic centres in the course of the synthesis. However, as you know from Chapter 41, making an enantiomerically pure compound like oseltamivir *must* involve a natural compound somewhere along the line. Diels–Alder reactions are catalysed by

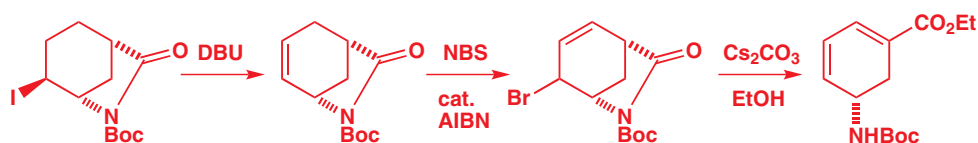
■ We illustrated a chiral auxiliary approach to asymmetric Diels–Alder reactions in Chapter 41, but it was clearly better to avoid the extra steps and recycling involved in large-scale synthesis with an auxiliary.

Lewis acids, and so by using a catalytic amount of the chiral Lewis acid (whose structure is evidently based on that of the CBS catalyst) it was possible to induce the cycloaddition to proceed enantioselectively.

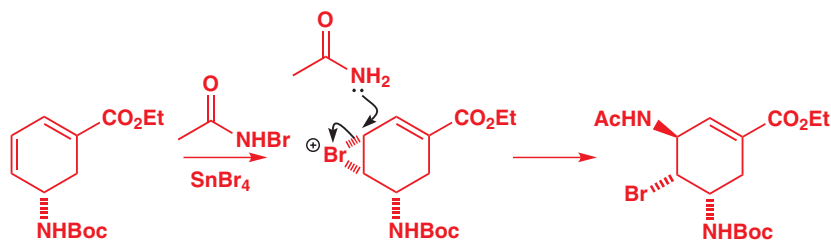
The product of the Diels–Alder reaction has the ester substituent in place, and the stereochemistry at the single chiral centre has to be used to control stereochemistry at new centres in the molecule. We discussed strategies for doing this in Chapters 32 and 33, and in this case the use of a tethered nucleophile (p. 847) allowed the first amino group to be introduced with the correct stereochemistry. Conversion of the ester to an amide followed by treatment with iodine induced in the nitrogen equivalent of an iodolactonization (an iodolactamization), placing the nitrogen *syn* to the ester and the iodide *trans*.



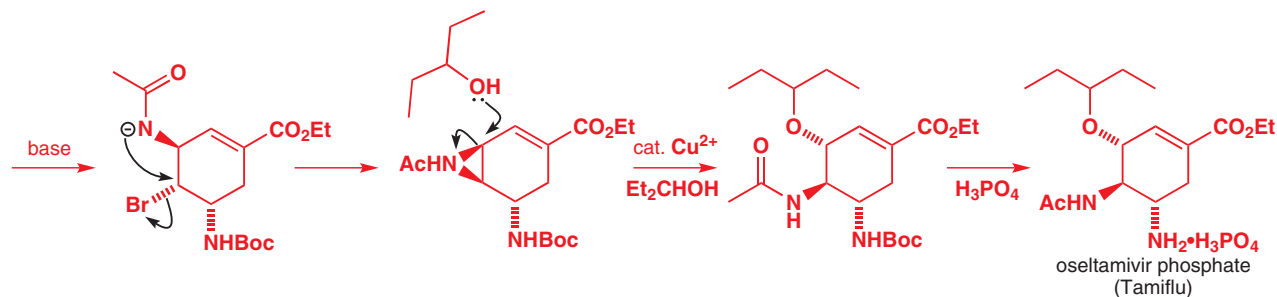
The next key intermediate is a diene, which is reached by an overall oxidation of the iodide: protection of nitrogen and elimination of the iodide gives the only possible alkene. Radical bromination with NBS followed by treatment with base in ethanol both hydrolyses the lactam and eliminates bromide to give the diene.



Now for the second nitrogen substituent. Bromination of the less electron deficient end of the diene with *N*-bromoacetamide in the presence of SnBr_4 leads to an intermediate bromonium ion which is opened by the acetamide by-product at the more reactive end adjacent to the alkene, giving a *trans* diaxial product.



Treatment with base leads to cyclization to an aziridine, and this time the ether is introduced by a copper-catalysed ring opening of the aziridine with 3-pentanol. Treatment with phosphoric acid removes the Boc protecting group and converts the product to oseltamivir phosphate.



Overall, Corey's route uses just 12 steps, and gives a yield of 30%—about double that of the route from shikimic acid. But much work remains to be done: several of the steps require

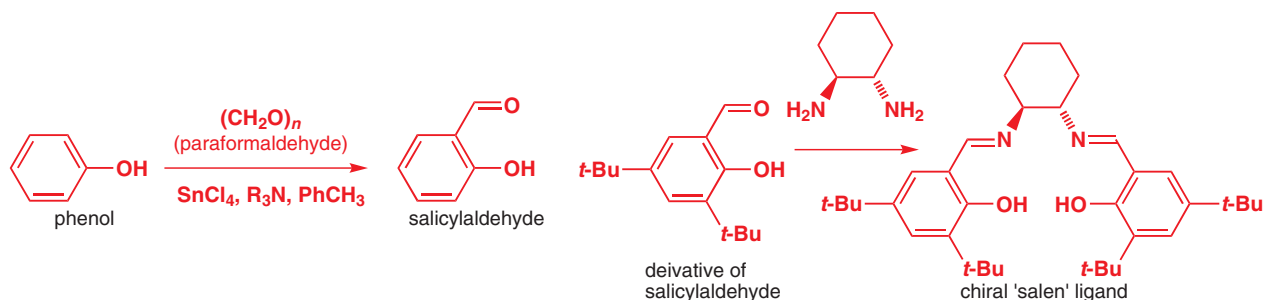
conditions or solvents (such as carbon tetrachloride) that are unsuitable for industrial use. Advances are still being made, with even shorter routes being reported since 2006. In some ways it would be best if this vital work were never made necessary, but across the world chemists are working in similar ways to relieve suffering, and potential suffering, caused by illness and disease.

The future of organic chemistry

Not all organic chemists can be involved in such exciting projects as the launching of a life-saving antiviral drug. Some most certainly have to be: resistant bacteria are fast catching up with our current range of antibiotics, and it is teams of organic chemists, in conjunction with biologists, who will be able to erect the next line of defence against these infections. But the chemistry used in such frontline projects is often the product of work by chemists in other institutions who had no idea that it would eventually be used to make a vital drug.

Take the millions of lives saved by the synthesis of indinavir, for example. This drug would not have been possible had not the Sharpless and Jacobsen asymmetric epoxidations, the catalytic asymmetric reduction, and the stereoselective enolate alkylation, along with many of the methods tried but not used in the final synthesis, been invented and developed by organic chemists in academic and industrial research laboratories. Some of the more famous names involved, like Sharpless, Jacobsen, and Noyori, invented new methods, while others modified and optimized those methods, and still others applied the methods to new types of molecules. Yet all built on the work of other chemists.

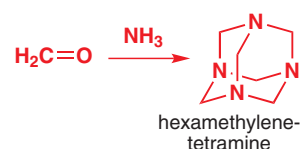
We can delve deeper into one of the steps in the indinavir synthesis. In 1980 Giovanni Casiraghi, a rather less famous chemist from the University of Parma, published a paper in the *Journal of the Chemical Society* about selective reactions between phenols and formaldehyde. He and his colleagues made the modest discovery that controlled reactions to give salicylaldehydes could be achieved in toluene with SnCl_4 as catalyst. The reaction is regioselective for the *ortho* isomer and the paper described the rather precise conditions needed to get a good yield.



The reaction was also successful for substituted salicylaldehydes. When Jacobsen came to develop his asymmetric epoxidation, he chose salens as his catalysts, partly because they could be made so easily from salicylaldehydes.

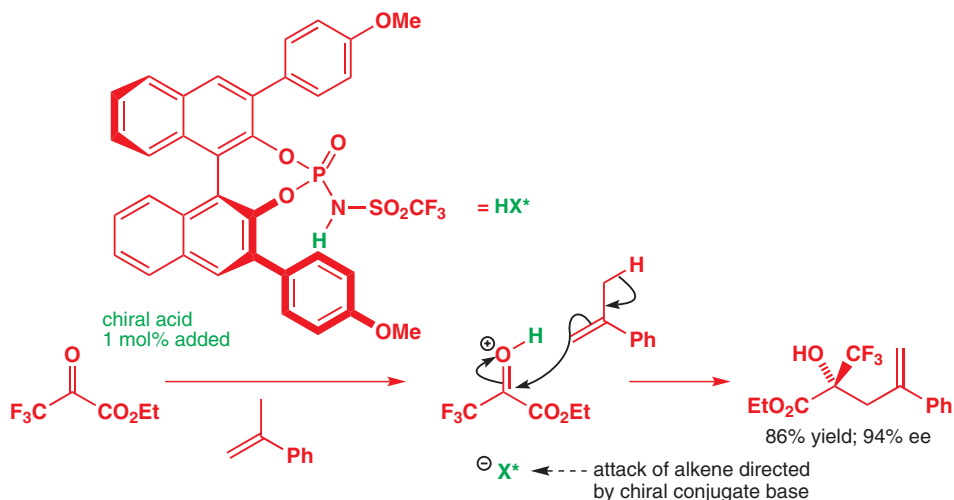
Jacobsen epoxidation turned out to be the best large-scale method for preparing the *cis*-aminoindanol for the synthesis of indinavir. This process is very much the cornerstone of the whole synthesis. It cannot have entered Casiraghi's wildest dreams that his work might someday be useful in a matter of life and death. Neither did his four co-workers nor Jacobsen's more numerous co-workers see clearly the future applications of their work. By its very nature it is impossible to predict the outcome or the applications of research. But one thing is certain: good research and exciting discoveries come from a thorough understanding of the fundamentals of organic chemistry.

When Jacobsen's epoxidation was fully described in 1998–99, the Casiraghi method was abandoned in favour of an even older method discovered in the 1930s by Duff. The remarkable Duff reaction uses hexamethylenetetramine, the oligomer of formaldehyde and ammonia, to provide the extra carbon atom. The now otherwise unknown Duff worked at Birmingham Technical College. Later in 1972, a William E. Smith, working in the GEC chemical laboratories



at Schenectady, New York, found how to make the Duff reaction more general and better yielding by using $\text{CF}_3\text{CO}_2\text{H}$ as catalyst. Even so, this method gives a lower yield than the Casiraghi method but it uses less toxic reagents (in particular it avoids stoichiometric tin) and is more suitable for large-scale work. When Duff was inventing his reaction or Smith was modifying the conditions, asymmetric synthesis was not even a gleam in anyone's eyes. It is impossible even for the inventor to predict whether a discovery is important or not.

Where is organic chemistry going next? As we write this chapter, advances are being made in reactions which would have seemed outlandish even just ten years ago. Work published in the years since 2005 has shown, for example, that many reactions of cations can be made to form single enantiomers of products even if they take place just in the vicinity of a chiral anion. Reactions such as the one below, from 2008, promise to revolutionize, yet again, some of the ways in which chemists make chiral compounds.



Interactive mechanism for catalytic enantioselective additions controlled by chiral anions

Finding drugs is a difficult job, and the number of new drugs launched each year is dropping as it becomes harder and more expensive to advance beyond existing treatments and as demands for more stringent safety rightly increase. But new drugs are made because...they can be made! What about all those classes of molecules which have never been made, simply because they have never been needed? Among them may well be molecules that will have all the specific attributes we want a potential drug to exhibit. Techniques known as diversity orientated synthesis are now addressing this idea—how to make and study great families of fundamentally different but potentially revolutionary molecules simply and efficiently. It's too early to tell, but the hope is that these techniques will provide breakthroughs in the fight against disease by finding completely new ways to attack their causes.

Nature is a superb synthetic chemist, and organic chemists have spent the last century exploring efficient ways of building molecular structures more efficiently than nature. Nature builds molecules a certain way because there is no alternative—molecules can be biosynthesized only if the enzymes to make them exist; enzymes are only made from the same 20 amino acids; amino acids are built into proteins by the same ribosome. The ribosome is the most complex and beautiful molecular structure in the known universe, but it can make only proteins. Chemists, with the periodic table, a supply of raw materials, a laboratory, and their ingenuity can make anything. Sometimes chemists use Nature's enzymes to do a job, or even force them to evolve to do a job better. By cloning useful enzymes in bacteria and forcing them to mutate, high-speed evolution can be induced, and enzymes can be created which do a job better, faster, or at a different temperature from their original 'wild type' ancestors.

More often chemists use reactions nature can never use—Rh, Ru, Pd, or phosphine ligands for that matter have never been exploited by any known biological process. What molecules chemists will make next, and how they make them, may determine the well-being of huge numbers of people in the future, but we may well not know it until then.

That future is yours as you continue your studies in organic chemistry beyond the scope of this book, and if you do you will want to read about modern work in more specialized areas. Your university library should have a selection of books on related topics we have only touched on, such as orbitals and chemical reactions, NMR spectroscopy, molecular modelling physical organic chemistry, photochemistry, enzyme mechanisms, biosynthesis, organometallic chemistry, asymmetric synthesis, supramolecular chemistry, and polymer and materials chemistry. This book will equip you with enough fundamental organic chemistry to explore these topics with understanding and enjoyment, and, perhaps, to discover what you want to do for the rest of your life. All of the chemists mentioned in this chapter and throughout the book began their careers as students of chemistry at universities somewhere in the world. You have the good fortune to study chemistry at a time when more is understood about the subject than ever before, when information is easier to retrieve than ever before, and when organic chemistry is more interrelated with other disciplines than ever before.

Further reading

For an informative overview of the most important drug molecules of the 20th century, see *Chemical and Engineering News*, 2005, Jun 20 edition.

Indinavir synthesis: I. W. Davies and P. J. Reider, *Chemistry and Industry (London)*, 1996, 412–15. G. Casiraghi, G. Casnati, G. Puglia, and G. Terenghi, *J. Chem. Soc., Perkin Trans. 1*, 1980, 1862–65.

Oseltamivir synthesis from shikimic acid: M. Federspiel and group, *Organic Process Research & Development* 1999, 3, 266–274. Corey

oseltamivir synthesis: Y-Y. Yeung, S. Hong, and E. J. Corey *J. Am. Chem. Soc.*, 2006, 128, 6310–631.

Chiral Brønsted acids: M. Rueping, B. J. Nachtsheim, W. Ieawsuwan, and I. Atodiresei, *Angew. Chem. Int. Ed.*, 2011, 50, 6706.

Diversity-orientated synthesis: D. Morton, S. Leach, C. Cordier, S. Warriner, and A. Nelson, *Angew. Chem. Int. Ed.*, 2009, 48, 104.

Check your understanding



To check that you have mastered the concepts presented in this chapter, attempt the problems that are available in the book's Online Resource Centre at <http://www.oxfordtextbooks.co.uk/orc/clayden2e/>